PMA Seminar Notes

Hansaem Park

August 5, 2026

About These Notes

These notes are intended to accompany the reading of Walter Rudin's *Principles of Mathematical Analysis*, rather than to serve as a self-contained replacement for the original text.

The statements of definitions, theorems, and exercises are generally reproduced from Rudin. Proofs and solutions are expanded, reorganized, or supplemented when additional detail seems useful. When Rudin's proof requires no further elaboration, it is omitted and only the statement is recorded.

Remarks, examples, and auxiliary results not appearing in the original text are occasionally added for clarification or later reference. Such additions are indicated when appropriate.

Contents

1 The Real and Complex Number Systems

1.1 Introduction

Example 1.1. It is well known that there is no rational p such that $p^2 = 2$.

Define $A \coloneqq \left\{p \in \mathbb{Q}^+ \mid p^2 < 2\right\}$ and $B \coloneqq \left\{p \in \mathbb{Q}^+ \mid p^2 > 2\right\}$. Then A contains no largest number and B contains no smallest.

Proof. It suffices to show that

$$\forall p \in A,\ \exists q \in A\, s.t.\, p < q$$

and

$$\forall p \in B,\ \exists q \in B\, s.t.\, q < p.$$

For every $p \in \mathbb{Q}^+$, take

$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}. \tag{1}$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}. \tag{2}$$

Hence,

$$\begin{aligned} p \in A &\implies p^2 - 2 < 0 \\ &\implies p < q \text{ and } q^2 < 2 && (\because (1), (2)) \\ &\implies q \in A \end{aligned}$$

and

$$\begin{aligned} p \in B &\implies p^2 - 2 > 0 \\ &\implies 0 < q < p \text{ and } q^2 > 2 && (\because (1), (2)) \\ &\implies q \in B. \end{aligned}$$

We have shown the desired result. □

1.2 Ordered Sets

Definition 1.2. Let S be a set. An *order* on S is a relation, denoted by $<$, with the following two properties:

(i) If $x \in S$ and $y \in S$ then one and only one of the statements

$$x < y, \qquad x = y, \qquad y < x$$

is true.

(ii) If $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Definition 1.3. An *ordered set* is a set S in which an order is defined.

Definition 1.4. Suppose S is an ordered set, and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for every $x \in E$, we say that E is *bounded above*, and call β an *upper bound* of E.

Lower bounds are defined in the same way (with $\geq$ in place of $\leq$).

Definition 1.5. Suppose S is an ordered set, $E \subset S$, and E is bounded above. Suppose there exists an $\alpha \in S$ with the following properties:

(i) α is an upper bound of E.

(ii) If $\gamma < \alpha$ then γ is not an upper bound of E.

Then α is called the *least upper bound of* E [that there is at most one such α is clear from (ii)] or the *supremum of* E, and we write

$$\alpha = \sup E.$$

The *greatest lower bound*, or *infimum*, of a set E which is bounded below is defined in the same manner: The statement

$$\alpha = \inf E$$

means that α is a lower bound of E and that no β with $\beta > \alpha$ is a lower bound of E.

Definition 1.6. An ordered set S is said to have the *least-upper-bound property* if the following is true:

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S.

Remark 1.7. The ordered set $\mathbb{Q}$ does not have the least-upper-bound property.

The following theorem shows that every ordered set with the least-upper-bound property also has the greatest-lower-bound property.

Theorem 1.8. *Suppose S is an ordered set with the least-upper-bound property, $B \subset S$, B is not empty, and B is bounded below. Let L be the set of all lower bounds of B. Then*

$$\alpha = \sup L$$

exists in S, and $\alpha = \inf B$.

In particular, $\inf B$ exists in S.

Proof. Since B is bounded below, L is nonempty. Let $x \in B$. Then $y \leq x$ for every $y \in L$. Thus, every $x \in B$ is an upper bound of L. Since L is a nonempty subset of S that is bounded above, by hypothesis,

$$\alpha = \sup L$$

exists in S.

If $\gamma < \alpha$ then γ is not an upper bound of L, hence $\gamma \notin B$. It follows that $\alpha \leq x$ for every $x \in B$. Thus, $\alpha \in L$.

If $\alpha < \beta$ then $\beta \notin L$, since α is an upper bound of L.

We conclude that α is a lower bound of B (since $\alpha \in L$) and if $\alpha < \beta$ then β is not a lower bound of B (since $\beta \notin L$), which implies $\alpha = \inf B$. □

Remark 1.9. Fix $x \in B$. Then $y \leq x$ for all $y \in L$. Thus, $\sup L \leq x$. Since the choice of $x \in B$ was arbitrary, we have

$$\sup L = \alpha \leq x \text{ for every } x \in B,$$

which implies α is a lower bound of B. Hence, $\alpha \in L$.

1.3 Fields

Definition 1.10. A *field* is a set F with two operations, called *addition* and *multiplication*, which satisfy the following so-called "field axioms" (A), (M), and (D):

(A) Axioms for addition

- (A1) If $x \in F$ and $y \in F$, then their sum $x + y$ is in F.
- (A2) Addition is commutative: $x + y = y + x$ for all $x, y \in F$.
- (A3) Addition is associative: $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.

(A4) F contains an element 0 such that $0 + x = x$ for every $x \in F$.

(A5) To every $x \in F$ corresponds an element $-x \in F$ such that

$$x + (-x) = 0.$$

(M) Axioms for multiplication

(M1) If $x \in F$ and $y \in F$, then their product xy is in F.

(M2) Multiplication is commutative: $xy = yx$ for all $x, y \in F$.

(M3) Multiplication is associative: $(xy)z = x(yz)$ for all $x, y, z \in F$.

(M4) F contains an element $1 \neq 0$ such that $1x = x$ for every $x \in F$.

(M5) If $x \in F$ and $x \neq 0$ then there exists an element $1/x \in F$ such that

$$x \cdot (1/x) = 1.$$

(D) The distributive law

$$x(y + z) = xy + xz$$

holds for all $x, y, z \in F$.

Definition 1.11. An *ordered field* is a *field* F which is also an *ordered set*, such that

(i) $x + y < x + z$ if $x, y, z \in F$ and $y < z$.

(ii) $xy > 0$ if $x \in F$, $y \in F$, $x > 0$, and $y > 0$.

1.4 The Real Field

Theorem 1.12. *There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.*

Theorem 1.13.

(a) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that

$$nx > y.$$

(b) If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists a $p \in \mathbb{Q}$ such that $x < p < y$.

Proof.

(a) Let $x, y \in \mathbb{R}$ and $x > 0$. Suppose to the contrary that $nx \leq y$ for each $n \in \mathbb{N}$ and define

$$A := \{nx \mid n \in \mathbb{N}\}.$$

Then A is nonempty and bounded above by y. Thus, the supremum of A exists in $\mathbb{R}$. Put $\alpha = \sup A$.

Since $x > 0$, we have $\alpha - x < \alpha$, hence, $\alpha - x$ is not an upper bound of A. Therefore, $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m + 1)x \in A$, which is a contradiction, since α is an upper bound of A.

(b) Since Rudin's proof of part (b) is already explicit, we omit it here.

□

Theorem 1.14. *For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$.*

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof. We omit the proof here. $\square$

Corollary. *If a and b are positive and real numbers and n is a positive integer, then*

$$(ab)^{\frac{1}{n}} = a^{\frac{1}{n}} b^{\frac{1}{n}}.$$

Lemma 1.15. [1]

(a) If $E \subset \mathbb{Z}$ has the infimum, then $\inf E \in E$.

(b) If $x \in \mathbb{R}$ then there exists an $n \in \mathbb{Z}$ such that

$$n \leq x < n + 1.$$

Proof.

(a) Let $\alpha = \inf E$. Suppose to the contrary that $\alpha \notin E$. Since $\alpha + 1$ is not a lower bound of E, choose an $x_0 \in E$ such that

$$\alpha < x_0 < \alpha + 1. \tag{3}$$

Since x_0 is not a lower bound of E, choose an $x_1 \in E$ again such that

$$\alpha < x_1 < x_0. \tag{4}$$

Subtracting x_0 from (4), we obtain $\alpha - x_0 < x_1 - x_0 < 0$. Since $-1 < \alpha - x_0$ from (3), it follows that $-1 < x_1 - x_0 < 0$. This contradicts the fact that $x_1 - x_0 \in \mathbb{Z}$. We conclude that $\alpha \in E$.

(b) Let $x \in \mathbb{R}$ and define $E \coloneqq \{n \in \mathbb{Z} \mid n > x\}$. By the Archimedean Property, E is nonempty. Since E is bounded below by x, the infimum of E exists in $\mathbb{R}$. Since $E \subset \mathbb{Z}$, $\inf E$ is the least element of E by (a). Thus, we have

$$\inf E - 1 \leq x < \inf E.$$

Putting $n = \inf E - 1 \in \mathbb{Z}$, we obtain the desired inequality.

$\square$

Remark 1.16 (Decimals). Let $x > 0$ be real. Let n_0 be the largest integer such that $n_0 \leq x$, by lemma 1.15. Having chosen $n_0, n_1, \ldots, n_{k-1}$, let n_k be the largest integer such that

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x.$$

Let E be the set of these numbers

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \qquad (k = 0, 1, 2, \ldots). \tag{5}$$

Notice that x is an upper bound of E and if $k \in \mathbb{N}$ then

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k + 1}{10^k}.$$

[1] This lemma has been added for convenience and does not appear in the original text.

Hence, we have

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < \frac{1}{10^k} \text{ for } k \in \mathbb{N}. \tag{6}$$

Suppose $y < x$. Then by the Archimedean Property, there exists a positive integer k such that $1/10^k < x - y$. From (6), we obtain

$$x - n_0 - \frac{n_1}{10} - \cdots - \frac{n_k}{10^k} < x - y,$$

hence

$$y < n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k}.$$

Therefore, $x = \sup E$. The decimal expansion of x is

$$n_0 \cdot n_1 n_2 n_3 \cdots . \tag{7}$$

Conversely, for any infinite decimal (7), the set E of numbers (5) is bounded above, and (7) is the decimal expansion of $\sup E$.

Exercises

Exercise 1. Let A be a nonempty set of real numbers which is bounded below. Let $-A$ be the set of all numbers $-x$, where $x \in A$. Prove that

$$\inf A = -\sup(-A).$$

Proof. Since A is bounded below, the infimum of A exists in $\mathbb{R}$. If $a \in A$ then $\inf A \leq a$. Hence,

$$-a \leq -\inf A \text{ for all } a \in A, \tag{8}$$

and it follows that $-A$ is bounded above. Thus, $\sup(-A)$ exists in $\mathbb{R}$. Since $-\inf A$ is an upper bound of $-A$ from (8), we have $\sup(-A) \leq -\inf A$, hence

$$\inf A \leq -\sup(-A). \tag{9}$$

Since $-a \leq \sup(-A)$ for every $a \in A$, we obtain $-\sup(-A) \leq a$ for all $a \in A$. Therefore, $-\sup(-A)$ is a lower bound of A and

$$-\sup(-A) \leq \inf A. \tag{10}$$

Combining (9) and (10), we conclude that $\inf A = -\sup(-A)$. $\square$

Exercise 2. Fix $b > 1$.

(a) If m, n, p, q are integers, $n > 0$, $q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If x is real, define $B(x)$ to be the set of all numbers b^t, where t is rational and $t \leq x$. Prove that

$$b^r = \sup B(r)$$

when r is rational. Hence it makes sense to define

$$b^x = \sup B(x)$$

for every real x.

(d) Prove that $b^{x+y} = b^x b^y$ for all real x and y.

Proof.

(a) Let $b^{mq} = b^{np} = x$. Since n and q are positive integers and $b > 0$, by Theorem 1.14, we have

$$b^m = x^{1/q} \text{ and } b^p = x^{1/n}.$$

It follows that

$$(b^m)^{1/n} = (x^{1/q})^{1/n} = (x^{1/n})^{1/q} = (b^p)^{1/q},$$

since $[(x^{1/q})^{1/n}]^{nq} = [(x^{1/n})^{1/q}]^{nq} = x$.

(b) Let m, n, p, q be integers and $n > 0$, $q > 0$ and let $r = m/n$ and $s = p/q$. Then by the Corollary to Theorem 1.14, we obtain

$$\begin{aligned} b^{r+s} &= b^{\frac{m}{n}+\frac{p}{q}} = b^{\frac{mq+np}{nq}} \\ &= \left(b^{mq+np}\right)^{\frac{1}{nq}} \\ &= \left(b^{mq}b^{np}\right)^{\frac{1}{nq}} \\ &= \left(b^{mq}\right)^{\frac{1}{nq}} \left(b^{np}\right)^{\frac{1}{nq}} \\ &= b^{\frac{mq}{nq}} b^{\frac{np}{nq}} = b^{\frac{m}{n}} b^{\frac{p}{q}} = b^r b^s. \end{aligned}$$

(c) Note that by Theorem 1.14, $b^r > 0$ if $b > 1$ and r is rational.

Let n be a positive integer and let $0 < a < 1$ be real. We first show that $0 < a^{1/n} < 1$.

Suppose to the contrary that $a^{1/n} \geq 1$. Then we obtain

$$1 \leq \left(a^{1/n}\right)^n = a < 1,$$

a contradiction. Therefore, if $0 < a < 1$ then $0 < a^{1/n} < 1$, where n is a positive integer.

Suppose r and s are rational. Let m, n, p, and q be integers and let n and q be positive. Then we write $r = m/n$, $s = p/q$.

Suppose $r < s$. Then

$$r - s = \frac{m}{n} - \frac{p}{q} = \frac{mq - np}{nq} < 0,$$

hence

$$0 < b^{mq-np} = \frac{1}{b^{np-mq}} < 1$$

and

$$0 < b^{r-s} = b^{\frac{mq-np}{nq}} = \left(b^{mq-np}\right)^{1/nq} < 1.$$

Thus, if $r < s$ then

$$b^r = b^{(r-s)+s} = b^{r-s}b^s < b^s.$$

If t is rational and $t \leq r$ then $b^t \leq b^r$. It follows that $B(r)$ is bounded above by b^r and $\sup B(r)$ exists in $\mathbb{R}$. Since b^r is an upper bound of $B(r)$,

$$\sup B(r) \leq b^r.$$

Since $b^r \in B(r)$,

$$b^r \leq \sup B(r).$$

We conclude that $b^r = \sup B(r)$.

(d) Let t_1 and t_2 be rational. If $t_1 \le x$ and $t_2 \le y$ then $t_1 + t_2 \le x + y$. Thus, we have

$$b^{t_1} b^{t_2} = b^{t_1+t_2} \le \sup B(x+y) = b^{x+y}.$$

Fix $t_1 \le x$. Then

$$b^{t_2} \le b^{x+y}/b^{t_1} \text{ for all } t_2 \le y.$$

Hence,

$$b^y = \sup B(y) \le b^{x+y}/b^{t_1}.$$

Since the choice of t_1 was arbitrary,

$$b^{t_1} \le b^{x+y}/b^y \text{ for all } t_1 \le x,$$

and it follows that

$$b^x = \sup B(x) \le b^{x+y}/b^y.$$

Therefore, we obtain

$$b^x b^y \le b^{x+y}. \tag{11}$$

Notice that

$$0 < b^{t_1} \le b^x \text{ for every } t_1 \le x$$

and

$$0 < b^{t_2} \le b^y \text{ for every } t_2 \le y.$$

Hence, for every $t_1 \le x$ and $t_2 \le y$,

$$b^{t_1+t_2} = b^{t_1} b^{t_2} \le b^x b^y.$$

Now, to obtain

$$b^{x+y} = \sup B(x+y) \le b^x b^y \tag{12}$$

from above inequality, we need the following statement:

$$b^x = \sup \left\{ b^t \mid t \in \mathbb{Q}, t < x \right\}.$$

Let us prove it. Notice that

$$b^n - 1 = (b-1)(b^{n-1} + \cdots + 1) \ge n(b-1)$$

for any positive integer n. Put $c = b^{1/n} > 1$. Then we have

$$b - 1 \ge n(b^{1/n} - 1).$$

Hence, if $t > 1$ and $n > (b-1)/(t-1)$ then

$$\frac{n(b^{1/n} - 1)}{t-1} \le \frac{b-1}{t-1} < n,$$

and it follows that

$$b^{1/n} < t.$$

Note that if $t > 1$ then $b^{1/n} < t$ for sufficiently large n.

Let $B_0(x) = \left\{ b^t \mid t \in \mathbb{Q}, t < x \right\}$. We need to show that b^x is the supremum of $B_0(x)$. It is clear that b^x is an upper bound of $B_0(x)$. Thus, it remains to prove that if $y < b^x$ then y is not an upper bound.

Suppose $0 < y < b^x$. Then $b^x/y > 1$ and it follows that

$$b^{1/n} < b^x/y$$

for sufficiently large n. Hence, we have

$$y < \frac{b^x}{b^{1/n}} < b^x.$$

By the density of rationals, choose a rational t such that $x - (1/n) < t < x$. Since $x < s < t + (1/n)$ for some rational s,

$$b^x \leq b^s < b^{t+(1/n)} = b^t b^{1/n}$$

and

$$y < \frac{b^x}{b^{1/n}} < b^t.$$

Thus, y is not an upper bound of $B_0(x)$ and therefore $b^x = \sup B_0(x)$.

Now, let t be a rational and let $t < x + y$. Pick $t_2 \in \mathbb{Q}$ such that $t - x < t_2 < y$ and set $t_1 = t - t_2$. Then for every $t \in \mathbb{Q}$ with $t < x + y$, there are rational $t_1 < x$ and $t_2 < y$ such that $t = t_1 + t_2 < x + y$, hence $b^t = b^{t_1+t_2} = b^{t_1} b^{t_2} \leq b^x b^y$. Thus, we obtain (12).

Combining (11) and (12), we conclude that

$$b^{x+y} = b^x b^y.$$

$\square$

Exercise 3. Prove that no order can be defined in the complex field that turns it into an ordered field.

Proof. Suppose to the contrary that there exists an order $<$ on $\mathbb{C}$ which turns $\mathbb{C}$ into an ordered field. Let $\mathbf{0}$ and $\mathbf{1}$ denote the additive and multiplicative identities of the complex field, respectively. Then $\mathbf{0} = (0, 0)$ and $\mathbf{1} = (1, 0)$.

Note that $i = (0, 1) \neq \mathbf{0}$. If $x \neq \mathbf{0}$ then $x^2 > \mathbf{0}$. It follows that

$$i^2 = (0, 1)(0, 1) = (-1, 0) > \mathbf{0}.$$

Since $(-1, 0) + (1, 0) = \mathbf{0}$, $(-1, 0)$ is the additive inverse of $(1, 0)$. Since $\mathbf{1} > \mathbf{0}$ and if $x > \mathbf{0}$ then $-x < \mathbf{0}$ in any ordered field, we obtain

$$(-1, 0) = -(1, 0) = -\mathbf{1} < \mathbf{0}.$$

This contradicts the Trichotomy Property. $\square$

2 Basic Topology

2.1 Finite, Countable, and Uncountable Sets

We omit this section here.

2.2 Metric Spaces

Definition 2.1. A set X, whose elements we shall call *points*, is said to be a *metric space* if with any two points p and q of X there is associated a real number $d(p, q)$, called the *distance* from p to q, such that

(a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;

(b) $d(p, q) = d(q, p)$;

(c) $d(p, q) \leq d(p, r) + d(r, q)$, for any $r \in X$.

Any function with these three properties is called a *distance function*, or a *metric*.

Example 2.2. $\mathbb{R}^k$ is a metric space.

Definition 2.3. By the *segment* (a, b) we mean the set of all real numbers x such that $a < x < b$.

By the *interval* $[a, b]$ we mean the set of all real numbers x such that $a \leq x \leq b$.

Occasionally we shall also encounter "half-open intervals" $[a, b)$ and $(a, b]$; the first consists of all x such that $a \leq x < b$, the second of all x such that $a < x \leq b$.

If $a_i < b_i$ for $i = 1, \ldots k$, the set of all points $\mathbf{x} = (x_1, \ldots, x_k)$ in $\mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ $(1 \leq i \leq k)$ is called a *k-cell.* Thus a 1-cell is an interval, a 2-cell is a rectangle, etc.

If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the *open* (or *closed*) *ball* B with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$ (or $|\mathbf{y} - \mathbf{x}| \leq r$).

We call a set $E \subset \mathbb{R}^k$ *convex* if

$$\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

Example 2.4. Balls and k-cells are convex.

Proof. If $|\mathbf{y} - \mathbf{x}| < r$, $|\mathbf{z} - \mathbf{x}| < r$, and $0 < \lambda < 1$, then

$$\begin{aligned}
|\lambda\mathbf{y} + (1 - \lambda)\mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + \lambda\mathbf{x} + (1 - \lambda)\mathbf{z} - \mathbf{x}| \\
&= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\
&\leq \lambda|\mathbf{y} - \mathbf{x}| + (1 - \lambda)|\mathbf{z} - \mathbf{x}| \\
&< \lambda r + (1 - \lambda)r = r.
\end{aligned}$$

Hence, open balls are convex. The same proof applies to closed balls.

Let $A \subset \mathbb{R}^k$ be a k-cell and let $\mathbf{x}, \mathbf{y} \in A$ and $0 < \lambda < 1$. We write $\mathbf{x} = (x_1, \ldots x_k)$, $\mathbf{y} = (y_1, \ldots, y_k)$ with

$$a_i < b_i \text{ and } x_i, y_i \in [a_i, b_i] \text{ for } i = 1, \ldots, k.$$

If $x, y \in [a, b]$ with $a < b$ and $0 < \lambda < 1$ then

$$\lambda a \leq \lambda x \leq \lambda b,$$

and

$$(1 - \lambda)a \leq (1 - \lambda)y \leq (1 - \lambda)b,$$

hence

$$a \le \lambda x + (1-\lambda)y \le b.$$

It follows that each $[a_i, b_i]$ $(1 \le i \le k)$ is convex.

Therefore, every k-cell is convex. □

Definition 2.5. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X.

(a) *A neighborhood* of p is a set $N_r(p)$ consisting of all q such that $d(p,q) < r$, for some $r > 0$. The number r is called the *radius* of $N_r(p)$.

(b) A point p is a *limit point* of the set E if *every* neighborhood of p contains a point $q \ne p$ such that $q \in E$.

(c) If $p \in E$ and p is not a limit point of E, then p is called an *isolated point* of E.

(d) E is *closed* if every limit point of E is a point of E.

(e) A point p is an *interior* point of E if there is a neighborhood N of p such that $N \subset E$.

(f) E is *open* if every point of E is an interior point of E.

(g) The *complement* of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.

(h) E is *perfect* if E is closed and if every point of E is a limit point of E.

(i) E is *bounded* if there is a real number M and a point $q \in X$ such that $d(p,q) < M$ for all $p \in E$.

(j) E is *dense in* X if every point of X is a limit point of E, or a point of E (or both).

Theorem 2.6. *Every neighborhood is an open set.*

Proof. Consider a neighborhood $E = N_r(p)$, and let q be any point of E. Then choose an $h > 0$ such that

$$d(p,q) = r - h.$$

If $s \in N_h(q)$ then

$$d(p,s) \le d(p,q) + d(q,s) < r - h + h = r,$$

and it follows that $N_h(q) \subset E$. Thus there is a neighborhood of q that is a subset of E. Therefore, every $q \in E$ is an interior point of E. We conclude that E is an open set. □

Theorem 2.7. *If p is a limit point of a set E, then every neighborhood of p contains infinitely many points of E.*

Proof. Suppose there is a neighborhood N of p which contains only a finite number of points of E. Let $q_1, \ldots, q_n$ be those points of $N \cap E$, which are distinct from p, and put

$$r = \min_{1 \le m \le n} d(p, q_m) > 0.$$

Then $d(p, q_m) \ge r$ for $m = 1, \ldots, n$ and hence all the points q_m are not in the neighborhood $N_r(p)$. Since

$$N_r(p) \subset N$$

and hence

$$N_r(p) \cap E \setminus \{p\} \subset N \cap E \setminus \{p\} = \{q_1, \ldots, q_n\} \subset E,$$

$N_r(p)$ contains no element $q \ne p$ in E (notice $N_r(p) \cap E \setminus \{p\}$ is empty). Thus, p is not a limit point of E. This is a contradiction. □

Corollary. *A finite point set has no limit points.*

Theorem 2.8. *Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α. Then*

$$\left(\bigcup_\alpha E_\alpha\right)^c = \bigcap_\alpha \left(E_\alpha^c\right). \tag{13}$$

Proof. We have

$$\begin{aligned} x \in \left(\bigcup_\alpha E_\alpha\right)^c &\implies x \notin \left(\bigcup_\alpha E_\alpha\right) \\ &\implies \forall \alpha, x \notin E_\alpha && (\because \text{negation of the definition}) \\ &\implies \forall \alpha, x \in E_\alpha^c \\ &\implies x \in \bigcap_\alpha E_\alpha^c. \end{aligned}$$

Hence $\left(\bigcup_\alpha E_\alpha\right)^c \subset \bigcap_\alpha \left(E_\alpha^c\right)$. Since

$$\begin{aligned} x \in \bigcap_\alpha \left(E_\alpha^c\right) &\implies \forall \alpha, x \in E_\alpha^c \\ &\implies \forall \alpha, x \notin E_\alpha \\ &\implies x \notin \bigcup_\alpha E_\alpha && (\because \text{negation of the definition}) \\ &\implies x \in \left(\bigcup_\alpha E_\alpha\right)^c, \end{aligned}$$

we obtain the reverse. This establishes the theorem. □

Theorem 2.9. *A set E is open if and only if its complement is closed.*

Proof. First, suppose E^c is closed. Choose $x \in E$. Then $x \notin E^c$, and x is not a limit point of E^c, since every limit point of E^c is a point of E^c (contrapositive). Hence there exists a neighborhood N of x such that $E^c \cap N$ is empty, that is, $N \subset E$. Thus E is open.

Next, suppose E is open. Let x be a limit point of E^c. Then every neighborhood of x contains a point of E^c, hence there is no neighborhood N of x such that $N \subset E$. Thus, x is not an interior point of E and therefore $x \notin E$, since E is open. It follows that if x is a limit point of E^c then $x \notin E$, hence $x \in E^c$. Therefore, E^c is closed. □

Corollary. *A set F is closed if and only if its complement is open.*

Theorem 2.10.

(a) For any collection $\{G_\alpha\}$ of open sets, $\bigcup_\alpha G_\alpha$ is open.

(b) For any collection $\{F_\alpha\}$ of closed sets, $\bigcap_\alpha F_\alpha$ is closed.

(c) For any finite collection $G_1, \dots G_n$ of open sets, $\bigcap_{i=1}^n G_i$ is open.

(d) For any finite collection $F_1, \dots, F_n$ of closed sets, $\bigcup_{i=1}^n F_i$ is closed.

Proof. Put $G = \bigcup_\alpha G_\alpha$. Then

$$\begin{aligned} x \in G &\implies \exists \alpha \, s.t. \, x \in G_\alpha \\ &\implies x \text{ is an interior point of } G_\alpha \\ &\implies \exists r_0 > 0 \, s.t. \, N_{r_0}(x) \subset G_\alpha \\ &\implies \exists r_0 > 0 \, s.t. \, N_{r_0}(x) \subset G. \end{aligned}$$

Hence, every $x \in G$ is an interior point of G and it follows that G is open.

(b) follows from (a), Theorem 2.8, and Theorem 2.9. We see that

$$\begin{aligned} F_\alpha \text{ is closed} &\implies F_\alpha^c \text{ is open.} && (\because \text{Theorem 2.9}) \\ &\implies \bigcup_\alpha F_\alpha^c \text{ is open.} && (\because \text{(a)}) \\ &\implies \left(\bigcap_\alpha F_\alpha \right)^c \text{ is open.} && (\because \text{Theorem 2.8}) \\ &\implies \bigcap_\alpha F_\alpha \text{ is closed.} && (\because \text{Theorem 2.9}) \end{aligned}$$

Since Rudin's proofs of parts (c) and (d) are already explicit, we omit them here. □

Example 2.11. For each $n \in \mathbb{N}$, define

$$G_n = \left(-\frac{1}{n}, \frac{1}{n} \right).$$

Then each G_n is an open subset of $\mathbb{R}^1$, but $\bigcap_{n=1}^\infty G_n = \{0\}$ is not. Similarly, let $F_n = G_n^c$ for every $n \in \mathbb{N}$. Then each F_n is closed, but $\bigcup_{n=1}^\infty F_n = \mathbb{R}^1 \setminus \{0\}$ is not.

Thus the intersection (respectively, union) of an infinite collection of open (respectively, closed) sets *need not be* open (respectively, closed). Hence the finiteness assumptions in parts (c) and (d) of Theorem 2.10 cannot be omitted.

Definition 2.12. If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X, then the *closure* of E is the set $\overline{E} = E \cup E'$.

Theorem 2.13. *If X is a metric space and $E \subset X$, then*

(a) $\overline{E}$ is closed,

(b) $E = \overline{E}$ if and only if E is closed,

(c) $\overline{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

By (a) and (c), $\overline{E}$ is the *smallest* closed subset of X that contains E.

Proof.

(a) If $p \in X$ and $p \notin \overline{E}$ then p is neither a point of E nor a limit point of E. Since p is not a limit point of E, there exists a neighborhood N of p such that for every point $q \in N$, $q = p$ or $q \notin E$. If $q = p$ then $q \notin E$, hence every point of N is not in E. Therefore N does not intersect E.

Suppose N contains a limit point l of E. Since N is open and $l \in N$, there exists a neighborhood N_l of l such that $N_l \subset N$. By this hypothesis we see that every neighborhood of l contains a point s in E. Thus $s \in N_l \subset N$ and this contradicts the fact that $N \cap E$ is empty. Therefore, N intersects neither E nor E'.

Since $(N \cap E) \cup (N \cap E') = \varnothing$, we obtain $N \cap (E \cup E') = N \cap \overline{E} = \varnothing$. Thus the complement of $\overline{E}$ contains N, hence it is open. This proves (a).

(b) If $E = \overline{E}$, (a) implies that E is closed. If E is closed, then $E' \subset E$, hence $\overline{E} = E$.

(c) We first show that if $E \subset F$ then $E' \subset F'$. Let $p \in E'$. If N is any neighborhood of p then there exists a point $q \neq p$ such that $q \in E \subset F$. Hence, every neighborhood N of p contains a point $q \neq p$ such that $q \in F$. Therefore, p is a limit point of F and $p \in F'$.

Let us prove (c) now. Suppose $F \subset X$ is closed and $E \subset F$. Since $E' \subset F'$ and $F' \subset F$, we see that $E' \subset F$, and it follows that $\overline{E} = E \cup E' \subset F$.

□

Theorem 2.14. *Let E be a nonempty set of real numbers which is bounded above. Let $y = \sup E$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.*

Proof. The result is immediate. □

Remark 2.15. Suppose $E \subset Y \subset X$, where X is a metric space. We say that E is *open relative to Y* if to each $p \in E$ there is associated an $r > 0$ such that $q \in E$ whenever $d(p, q) < r$ and $q \in Y$.

Theorem 2.16. *Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X.*

2.3 Compact Sets

Definition 2.17. By an *open cover* of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \bigcup_\alpha G_\alpha$.

Definition 2.18. A subset K of a metric space X is said to be *compact* if every open cover of K contains a *finite* subcover.

More explicitly, the requirement is that if $\{G_\alpha\}$ is an open cover of K, then there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}.$$

Theorem 2.19. *Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y.*

Proof. Suppose K is compact relative to X, and let $\{V_\alpha\}$ be a collection of sets, open relative to Y, such that $K \subset \bigcup_\alpha V_\alpha$. By Theorem 2.16, there are sets G_α, open relative to X, such that $V_\alpha = Y \cap G_\alpha$, for all α; and since K is compact relative to X, we have

$$K \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n} \tag{14}$$

for some choice of finitely many indices $\alpha_1, \dots, \alpha_n$. Since $K \subset Y$, we have $K \cap Y = K$. Hence (14) implies

$$K = K \cap Y \subset (G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}) \cap Y = V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}. \tag{15}$$

Since $\{V_\alpha\}$ was an arbitrary open cover (relative to Y) of K, (15) proves that K is compact relative to Y.

Conversely, suppose K is compact relative to Y, let $\{G_\alpha\}$ be a collection of open subsets of X which covers K, and put $V_\alpha = Y \cap G_\alpha$. Then (15) will hold for some choice of $\alpha_1, \dots, \alpha_n$; and since $V_\alpha \subset G_\alpha$, (15) implies (14).

This completes the proof. □

Theorem 2.20. *Compact subsets of metric spaces are closed.*

Proof. Let K be a compact subset of a metric space X. We shall prove that the complement of K is an open subset of X.

Suppose $p \in X$, $p \notin K$. If $q \in K$, let V_q and W_q be neighborhoods of p and q, respectively, of radius less than $\frac{1}{2}d(p,q)$. Notice that the collection of the neighborhoods W_q of the points $q \in K$ is an open cover of K. Since K is compact, there are finitely many points $q_1, \dots, q_n$ in K such that

$$K \subset W_{q_1} \cup \cdots \cup W_{q_n} = W.$$

If $p, q \in X$ then $d(p,q) \le d(p,x) + d(x,q)$ for any $x \in X$. Hence, at least one of $d(p,x)$ and $d(x,q)$ is not less than $\frac{1}{2}d(p,q)$. Thus, V_{q_i} and W_{q_i} are disjoint for $i = 1, \dots, n$. It follows that if $V = V_{q_1} \cap \cdots \cap V_{q_n}$, then V is a neighborhood of p which does not intersect W. Hence $V \subset K^c$, so that p is an interior point of K^c. The theorem follows. □

Theorem 2.21. *Closed subsets of compact sets are compact.*

Corollary. *If F is closed and K is compact, then $F \cap K$ is compact.*

Proof. Theorems 2.10(b) and 2.20 show that $F \cap K$ is closed; since $F \cap K \subset K$, Theorem 2.21 shows that $F \cap K$ is compact. □

Theorem 2.22. *If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then $\bigcap K_\alpha$ is nonempty.*

Proof. Fix a member K_1 of $\{K_\alpha\}$ and put $G_\alpha = K_\alpha^c$. Assume that no point of K_1 belongs to every K_α. Then for any $x \in K_1$, there exists an α such that $x \in G_\alpha$. Since every K_α is closed by Theorem 2.20, every G_α is open. Hence, the sets G_α form an open cover of K_1; and since K_1 is compact, there are finitely many indices $\alpha_1, \dots, \alpha_n$ such that $K_1 \subset G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$. But this means that

$$K_1 \cap K_{\alpha_1} \cap \cdots \cap K_{\alpha_n}$$

is empty, in contradiction to our hypothesis. □

Corollary. *If $\{K_n\}$ is a sequence of nonempty compact sets such that $K_n \supset K_{n+1}$ $(n = 1, 2, 3, \dots)$, then $\bigcap_1^\infty K_n$ is not empty.*

Theorem 2.23. *If E is an infinite subset of a compact set K, then E has a limit point in K.*

Theorem 2.24. *If $\{I_n\}$ is a sequence of intervals in $\mathbb{R}^1$, such that $I_n \supset I_{n+1}$ $(n = 1, 2, 3, \dots)$, then $\bigcap_1^\infty I_n$ is not empty.*

Theorem 2.25. *Let k be a positive integer. If $\{I_n\}$ is a sequence of k-cells such that $I_n \supset I_{n+1}$ $(n = 1, 2, 3, \dots)$, then $\bigcap_1^\infty I_n$ is not empty.*

Theorem 2.26. *Every k-cell is compact.*

Theorem 2.27. *If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two:*

(a) *E is closed and bounded.*

(b) *E is compact.*

(c) *Every infinite subset of E has a limit point in E.*

Proof. If (a) holds, then $E \subset I$ for some k-cell I, and (b) follows from Theorems 2.26 and 2.21. Theorem 2.23 shows that (b) implies (c). It remains to be shown that (c) implies (a).

If E is not bounded, then E contains points $\mathbf{x}_n$ with

$$|\mathbf{x}_n| > n \qquad (n = 1, 2, 3, \ldots).$$

The set S consisting of these points $\mathbf{x}_n$ is infinite and has no limit point in $\mathbb{R}^k$, hence none in E. For if $p \in \mathbb{R}^k$ were a limit point of S, then there would exist a positive integer n such that $|p| < n$ by the Archimedean Property. Put $r = n - |p|$. Then the neighborhood $N_r(p)$ of p contains at most $n-1$ points of S, which contradicts Theorem 2.7. Thus (c) implies that E is bounded.

If E is not closed, then there is a point $\mathbf{x}_0 \in \mathbb{R}^k$ which is a limit point of E but not a point of E. For $n = 1, 2, 3, \ldots$, there are points $\mathbf{x}_n \in E$ such that $|\mathbf{x}_n - \mathbf{x}_0| < 1/n$. Let S be the set of these points $\mathbf{x}_n$. Then S is infinite (otherwise $|\mathbf{x}_n - \mathbf{x}_0|$ would have a constant positive value, for infinitely many n), S has $\mathbf{x}_0$ as a limit point, and S has no other limit point in $\mathbb{R}^k$. For if $\mathbf{y} \in \mathbb{R}^k$, $\mathbf{y} \neq \mathbf{x}_0$, then

$$\begin{aligned} |\mathbf{x}_n - \mathbf{y}| &\geq |\mathbf{x}_0 - \mathbf{y}| - |\mathbf{x}_n - \mathbf{x}_0| \\ &\geq |\mathbf{x}_0 - \mathbf{y}| - \frac{1}{n} \geq \frac{1}{2}|\mathbf{x}_0 - \mathbf{y}| \end{aligned}$$

for all but finitely many n; this shows that $\mathbf{y}$ is not a limit point of S (Theorem 2.7).

Thus S has no limit point in E; hence E must be closed if (c) holds. □

Theorem 2.28 (Weierstrass). *Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.*

2.4 Perfect Sets

We omit this section.

2.5 Connected Sets

Definition 2.29. Two subsets A and B of a metric space X are said to be *separated* if both $A \cap \overline{B}$ and $\overline{A} \cap B$ are empty.

A set $E \subset X$ is said to be *connected* if E is *not* a union of two nonempty separated sets.

Theorem 2.30. *A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property: If $x \in E$, $y \in E$, and $x < z < y$, then $z \in E$.*

Proof. If there exist $x \in E$, $y \in E$, and some $z \in (x, y)$ such that $z \notin E$, then $E = A_z \cup B_z$ where

$$A_z = E \cap (-\infty, z), \qquad B_z = E \cap (z, \infty).$$

Since $x \in A_z$ and $y \in B_z$, A_z and B_z are nonempty. Since $A_z \subset (-\infty, z)$, and $B_z \subset (z, \infty)$, we have

$$\overline{A_z} \subset (-\infty, z] \text{ and } \overline{B_z} \subset [z, \infty).$$

Thus, both $\overline{A_z} \cap B_z$ and $A_z \cap \overline{B_z}$ are empty, and it follows that A_z and B_z are separated. Hence E is not connected.

To prove the converse, suppose E is not connected. Then there are nonempty separated sets A and B such that $A \cup B = E$. Pick $x \in A$, $y \in B$, and assume (without loss of generality) that $x < y$. Define

$$z = \sup(A \cap [x, y]).$$

By Theorem 2.14, $z \in \overline{A \cap [x, y]} \subset \overline{A}$; hence $z \notin B$. Since $A \cap [x, y] \subset [x, y]$, we have

$$z = \sup(A \cap [x, y]) \leq \sup[x, y] = y,$$

and it follows that $x \le z < y$.

If $z \notin A$, it follows that $x < z < y$ and $z \notin E$.

If $z \in A$, then $z \notin \overline{B}$, hence z is not a limit point of B. Thus, there exists an $r > 0$ such that $(z-r, z+r) \cap B = \varnothing$. Since $y \in B$, we have $y \le z - r$ or $y \ge z + r$. Since $z < y$, it follows that

$$z < z + \frac{r}{2} < z + r \le y.$$

Hence there exists z_1 such that $z < z_1 < y$ and $z_1 \notin B$ (take $z_1 = z + r/2$). If $z_1 \in A$ then $z_1 \in A \cap [x, y]$, hence $z_1 = z + r/2 \le z$, a contradiction. Therefore, $x < z_1 < y$ and $z_1 \notin E$. $\square$

Exercises

Exercise 4. Let E' be the set of all limit points of a set E. Prove that E' is closed. Prove that E and $\overline{E}$ have the same limit points. (Recall that $\overline{E} = E \cup E'$.) Do E and E' always have the same limit points?

Proof. Let p be a limit point of E' and let N be any neighborhood of p. Then N contains a point $q \neq p$ such that $q \in E'$. Since N is open, there exists a neighborhood N_q of q such that $N_q \subset N$. Without loss of generality, assume that N_q does not contain p. For if $p \in N_q$ then we take $r = d(p, q) > 0$, hence $p \notin N_r(q) \subset N_q \subset N$.

Recall that q is a limit point of E ($q \in E'$), we see that N_q contains a point $s \neq q$ such that $s \in E$. since $p \notin N_q$, we have $p \neq s$. Therefore, $N \supset N_q$ contains a point $s \neq p$ such that $s \in E$. Since N was arbitrary, p is a limit point of E and $p \in E'$.

Since the choice of p was arbitrary, every limit point of E' is a point of E'. Thus, E' is closed.

Next, let us prove that E and $\overline{E}$ have the same limit points. Suppose p is a limit point of E. Then every neighborhood of p contains a point $q \neq p$ such that $q \in E \subset E \cup E' = \overline{E}$. Thus p is a limit point of $\overline{E}$.

Conversely, suppose p is a limit point of $\overline{E}$. Then every neighborhood N of p contains a point $q \neq p$ such that $q \in \overline{E}$. Note that $q \in E$ or $q \in E'$.

If $q \in E$ then we are done. Otherwise, suppose $q \in E'$. Since N is open, there is a neighborhood N_q of q such that $N_q \subset N$. Without loss of generality, assume N_q does not contain p. Since q is a limit point of E, N_q contains a point $s \neq q$ such that $s \in E$. Hence, N contains a point $s \neq p$ (since $p \notin N_q$) such that $s \in E$. Since N was arbitrary, p is a limit point of E.

We conclude that E and $\overline{E}$ have the same limit points.

Finally, let $E = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$. It is clear that $E' = \{0\}$ but there is no limit point of E'. Hence it is possible that E and E' have different limit points. $\square$

Exercise 5. Let $A_1, A_2, A_3, \ldots$ be subsets of a metric space.

(a) If $B_n = \bigcup_{i=1}^{n} A_i$, prove that $\overline{B_n} = \bigcup_{i=1}^{n} \overline{A_i}$, for $n = 1, 2, 3, \ldots$.

(b) If $B = \bigcup_{i=1}^{\infty} A_i$, prove that $\overline{B} \supset \bigcup_{i=1}^{\infty} \overline{A_i}$.

Show, by an example, that this inclusion can be proper.

Proof. To prove (a), by mathematical induction, it suffices to prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$. We need to prove that $(A \cup B)' = A' \cup B'$.

Suppose $p \in (A \cup B)'$. Assume, for contradiction, that p is neither a limit point of A nor B. Then there are neighborhoods N_1 and N_2 of p such that $N_1 \cap (A \setminus \{p\})$ and $N_2 \cap (B \setminus \{p\})$ are empty, respectively.

Let $N = N_1 \cap N_2$. Then we have

$$\begin{aligned}[N \cap (A \setminus \{p\})] \cup [N \cap (B \setminus \{p\})] &= N \cap [(A \setminus \{p\}) \cup (B \setminus \{p\})] \\ &= N \cap [(A \cup B) \setminus \{p\}] = \varnothing.\end{aligned}$$

This contradicts our assumption. Therefore, $(A \cup B)' \subset A' \cup B'$.

Conversely, suppose $p \in A' \cup B'$. Then $p \in A'$ or $p \in B'$. If $p \in A'$, then every neighborhood of p contains a point $q \neq p$ such that $q \in A \subset A \cup B$, hence $p \in (A \cup B)'$. Similarly, if $p \in B'$ then p is a limit point of $A \cup B$, hence $p \in (A \cup B)'$. This proves $A' \cup B' \subset (A \cup B)'$.

We have shown that $(A \cup B)' \subset A' \cup B'$ and $A' \cup B' \subset (A \cup B)'$. Hence $(A \cup B)' = A' \cup B'$ and it follows that

$$\begin{aligned}\overline{A \cup B} &= (A \cup B) \cup (A \cup B)' = (A \cup B) \cup (A' \cup B') \\ &= (A \cup A') \cup (B \cup B') = \overline{A} \cup \overline{B}.\end{aligned}$$

This is the desired result.

Next, let us prove (b). Suppose $p \in \bigcup_{i=1}^{\infty} \overline{A_i}$. Then $p \in \overline{A_i}$ for some positive integer i. If $p \in A_i$ then $p \in B$. If $p \in A_i'$ then every neighborhood of p contains a point $q \neq p$ such that $q \in A_i \subset \bigcup_{i=1}^{\infty} A_i = B$. Therefore, p is a limit point of B and $p \in B'$.

Hence, if $p \in \bigcup_{i=1}^{\infty} \overline{A_i}$ then $p \in B$ or $p \in B'$. This proves $\bigcup_{i=1}^{\infty} \overline{A_i} \subset \overline{B}$.

Finally, we show that this inclusion can be proper. For each $n \in \mathbb{N}$, define

$$A_n = \left\{ x \in \mathbb{R} \mid \frac{1}{n} < x \right\}.$$

Then we see that B is the set of positive real numbers. Observe that 0 is not a limit point of A_n and $0 \notin A_n$ for $n = 1, 2, 3, \ldots$. It follows that $0 \notin \bigcup_{i=1}^{\infty} \overline{A_i}$. But it is clear that $0 \in \overline{B}$. □

Exercise 6. Is every point of every open set $E \subset \mathbb{R}^2$ a limit point of E? Answer the same question for closed sets in $\mathbb{R}^2$.

Proof. Suppose $p \in E$ is not a limit point of E. Then there exists a neighborhood N' of p such that $N' \cap E = \{p\}$. Since E is open, choose a neighborhood N'' of p such that $N'' \subset E$. Then we have

$$N = (N'' \cap N') \subset (E \cap N') = \{p\}.$$

Since $p \in N$, it follows that $N = \{p\}$. Since N is an intersection of two neighborhoods of the same point p in $\mathbb{R}^2$, N is a neighborhood of p. But in $\mathbb{R}^2$, every neighborhood is infinite. This is a contradiction.

Next, let $F = \{\mathbf{x}\} \subset \mathbb{R}^2$. It is clear that F is a closed subset of $\mathbb{R}^2$ but $\mathbf{x} \in F$ is not a limit point of F. □

Exercise 7. Let $K \subset \mathbb{R}^1$ consist of 0 and the numbers $1/n$, for $n = 1, 2, 3, \ldots$. Prove that K is compact directly from the definition (without using the Heine-Borel theorem).

Proof. Let $\{G_\alpha\}$ be an open cover of K. Since $0 \in K$, there exists an α_0 such that $0 \in G_{\alpha_0}$. Since every G_α is open, choose a neighborhood N of 0 with radius r such that $N \subset G_{\alpha_0}$.

If $r > 1$ then $K \subset G_{\alpha_0}$ hence we are done.

If $0 < r \leq 1$ then the set

$$E = \{1/n \mid n \in \mathbb{N}, r \leq 1/n\}$$

is finite by the Archimedean Property. Let $x_1, \ldots, x_k$ be the points of E. Since $x_i \in K \subset \bigcup_\alpha G_\alpha$ $(1 \leq i \leq k)$, there exists α_i such that $x_i \in G_{\alpha_i}$ for $i = 1, \ldots, k$. Thus, we have finitely many indices $\alpha_0, \alpha_1, \ldots, \alpha_k$ such that

$$K \subset G_{\alpha_0} \cup G_{\alpha_1} \cup \cdots \cup G_{\alpha_k}.$$

We conclude that K is compact. □

Exercise 8. Give an example of an open cover of the segment $(0,1)$ which has no finite subcover.

Proof. For each $n \in \mathbb{N}$, let

$$G_n = \left\{ x \in \mathbb{R} \mid 0 < x < 1 - \frac{1}{n} \right\}.$$

If $0 < x < 1$, then there exists a positive integer n such that $1 < n(1-x)$, hence $0 < x < 1 - \frac{1}{n}$ so that $x \in G_n$. It follows that the collection $\{G_n\}_{n\in\mathbb{N}}$ is an open cover of $(0,1)$. But it is clear that any finite subcollection of $\{G_n\}_{n\in\mathbb{N}}$ cannot cover $(0,1)$. □

3 Numerical Sequences and Series

3.1 Convergent Sequences

Definition 3.1. A sequence $\{p_n\}$ in a metric space X is said to *converge* if there is a point $p \in X$ with the following property: For every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies that $d(p_n, p) < \varepsilon$. (Here d denotes the distance in X.)

In this case we also say that $\{p_n\}$ converges to p, or that p is the limit of $\{p_n\}$ [see Theorem 3.2(b)], and we write $\{p_n\} \to p$, or

$$\lim_{n\to\infty} p_n = p.$$

If $\{p_n\}$ does not converge, it is said to *diverge*.

We recall that the set of all points p_n $(n = 1, 2, 3, \ldots)$ is the *range* of $\{p_n\}$. The range of a sequence may be a finite set, or it may be infinite. The sequence $\{p_n\}$ is said to be *bounded* if its range is bounded.

Theorem 3.2. *Let $\{p_n\}$ be a sequence in a metric space X.*

(a) $\{p_n\}$ converges to $p \in X$ if and only if every neighborhood of p contains p_n for all but finitely many n.

(b) If $p \in X$, $p' \in X$, and if $\{p_n\}$ converges to p and to p', then $p' = p$.

(c) If $\{p_n\}$ converges, then $\{p_n\}$ is bounded.

(d) If $E \subset X$ and if p is a limit point of E, then there is a sequence $\{p_n\}$ in E such that $p = \lim\limits_{n\to\infty} p_n$.

Proof. Since Rudin's proofs of parts (a), (b), and (c) are already explicit, we omit them here.

We prove part (d). Since p is a limit point of E, for $n = 1, 2, 3, \ldots$, choose a point $p_n \in E$ such that $d(p_n, p) < 1/n$. Given $\varepsilon > 0$, there exists a positive integer N such that $N\varepsilon > 1$. Then $n \geq N$ implies

$$d(p_n, p) < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore, $\{p_n\}$ converges to p. $\square$

Theorem 3.3. *Suppose $\{s_n\}$, $\{t_n\}$ are complex sequences, and $\lim_{n\to\infty} s_n = s$, $\lim_{n\to\infty} t_n = t$. Then*

(a) $\lim\limits_{n\to\infty} (s_n + t_n) = s + t$;

(b) $\lim\limits_{n\to\infty} cs_n = cs$, $\lim\limits_{n\to\infty} (c + s_n) = c + s$, for any number c;

(c) $\lim\limits_{n\to\infty} s_n t_n = st$;

(d) $\lim\limits_{n\to\infty} \dfrac{1}{s_n} = \dfrac{1}{s}$, provided $s_n \neq 0$ $(n = 1, 2, 3, \ldots)$, and $s \neq 0$.

Theorem 3.4.

(a) Suppose $\mathbf{x}_n \in \mathbb{R}^k$ $(n = 1, 2, 3, \ldots)$ and

$$\mathbf{x}_n = (\alpha_{1,n}, \ldots, \alpha_{k,n}).$$

Then $\{\mathbf{x}_n\}$ converges to $\mathbf{x} = (\alpha_1, \ldots, \alpha_k)$ if and only if

$$\lim_{n\to\infty} \alpha_{j,n} = \alpha_j \qquad (1 \leq j \leq k). \tag{16}$$

(b) Suppose $\{\mathbf{x}_n\}$, $\{\mathbf{y}_n\}$ are sequences in $\mathbb{R}^k$, $\{\beta_n\}$ is a sequence of real numbers, and $\mathbf{x}_n \to \mathbf{x}$, $\mathbf{y}_n \to \mathbf{y}$, $\beta_n \to \beta$. Then

$$\lim_{n\to\infty} (\mathbf{x}_n + \mathbf{y}_n) = \mathbf{x} + \mathbf{y}, \qquad \lim_{n\to\infty} \mathbf{x}_n \cdot \mathbf{y}_n = \mathbf{x} \cdot \mathbf{y}, \qquad \lim_{n\to\infty} \beta_n \mathbf{x}_n = \beta\mathbf{x}.$$

3.2 Subsequences

Definition 3.5. Given a sequence $\{p_n\}$, consider a sequence $\{n_k\}$ of positive integers, such that $n_1 < n_2 < n_3 < \cdots$. Then the sequence $\{p_{n_i}\}$ is called a *subsequence* of $\{p_n\}$. If $\{p_{n_i}\}$ converges, its limit is called a *subsequential limit* of $\{p_n\}$.

A sequence $\{p_n\}$ converges to p if and only if every subsequence of $\{p_n\}$ converges to p.

To prove that, suppose $\{p_n\}$ converges to p and let $\{p_{n_i}\}$ be a subsequence of $\{p_n\}$. Given $\varepsilon > 0$, there exists a positive integer N such that $d(p_n, p) < \varepsilon$ for $n \geq N$. Since $n_i \geq i$ for $i = 1, 2, 3, \ldots$, $i \geq N$ implies $n_i \geq N$ and therefore $d(p_{n_i}, p) < \varepsilon$. Hence $\{p_{n_i}\}$ converges to p.

The converse is trivial.

Theorem 3.6.

(a) If $\{p_n\}$ is a sequence in a compact metric space X, then some subsequence of $\{p_n\}$ converges to a point of X.

(b) Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Proof.

(a) Let E be a range of $\{p_n\}$. If E is finite then there are infinitely many terms such that

$$p_{n_1} = p_{n_2} = p_{n_3} = \cdots = p.$$

Without loss of generality, we may assume that $n_1 < n_2 < n_3 < \cdots$. It is clear that $\{p_{n_i}\}$ converges to p.

If E is infinite, Theorem 2.23 shows that E has a limit point $p \in X$. Choose n_1 so that $d(p, p_{n_1}) < 1$. Having chosen $n_1, \ldots, n_{i-1}$, we see from Theorem 2.7 that there is an integer $n_i > n_{i-1}$ such that $d(p, p_{n_i}) < 1/i$. Then $\{p_{n_i}\}$ converges to p.

(b) This follows from (a) and the Heine–Borel Theorem.

□

Theorem 3.7. *The subsequential limits of a sequence $\{p_n\}$ in a metric space X form a closed subset of X.*